Yong Li Neo

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RESEARCH INTERESTS

Machine learning for biology; graph and sheaf neural networks; gene regulatory networks (GRNs); cell-cell interactions; cell lineage reconstruction.

EDUCATION

Georgia Institute of Technology — Atlanta, GA, USA
B.S. in Computer Science, Minor in Mathematics — Expected December 2027
GPA: 4.0/4.0

RESEARCH EXPERIENCE

Zhang CompBio Group, Georgia Institute of Technology — Atlanta, GA, USA
Undergraduate Research Assistant, advised by Prof. Xiuwei Zhang — January 2026 – Present

- Research on gene regulatory networks, cell-cell interaction, and cell lineage reconstruction.
- Develop graph and sheaf neural network models to uncover cell-cell interactions in spatial and multi-modal single-cell data.
- Funded by the President's Undergraduate Research Award (PURA Salary Award), April 2026.
- Present progress and discuss results with peers at weekly group meetings.

PUBLICATIONS

Peer-reviewed conference papers (IEEE)

[1] **Y. L. Neo**, "A Review of State-of-the-Art Machine Algorithms on Classifying Depressive Disorder EEG Signals," *ICAICA*, Dalian, China, 2024.

[2] Z. Y. Lim and **Y. L. Neo**, "Confused vs Non-Confused Electroencephalography Signal Classification Using Deep Learning Algorithm," *I2CACIS*, Shah Alam, Malaysia, 2023.

[3] Z. Y. Lim and **Y. L. Neo**, "Epileptic Seizure Electroencephalography Signal Classification Using Machine Learning Algorithm," *CSCI*, Las Vegas, NV, USA, 2022.

RESEARCH PROJECTS

Depression Detection from EEG Signals — Publication [1]

- Built ML models that identified depressed patients with 80% accuracy, in collaboration with a local mental health clinic.
- Benchmarked Elastic Net, SVM, Random Forest, XGBoost, and CatBoost for medical signal processing.

Confusion Detection from EEG Signals — Publication [2]

- Compared CNN-biLSTM, CNN-biGRU, Bi-RNN, and Bi-GRU architectures; the best model reached 76.7% accuracy.

Epileptic Seizure Detection from EEG Signals — Publication [3]

- Developed and validated models with Extra Trees, Naïve Bayes, SGD, and Logistic Regression, reaching 96.6% accuracy.

HONORS, AWARDS & FELLOWSHIPS

President's Undergraduate Research Award (PURA Salary Award), Georgia Tech — April 2026
International Olympiad in Informatics (IOI), Participation Award — 2023
Asia Pacific Olympiad in Informatics (APIO), Participation Award — 2023, 2024
Kolej Tuanku Ja'afar Scholarship, Recipient

TEACHING, MENTORING & LEADERSHIP

Malaysian Informatics and Programming Society (MIPS) Malaysia
National Trainer & Problem Setter December 2023 – May 2025

- Taught lectures and weekly online workshops for the top 50 national finalists.
- Designed national Informatics Olympiad problems on dynamic programming, binary search, and sorting.

Robotics Club & Tech Club, Kolej Tuanku Ja'afar (KTJ) Malaysia
Vice President & Head of Teaching March 2024 – May 2025

- Led 30+ weekly meetings and mentored 60 junior members in Arduino and C++.

OTHER PROJECTS

Competitive Programming for Beginners (C++ book), Author December 2023 – June 2024

- Sold 100 copies and donated $3,000 in profits to charity; featured in *The Star*, a national newspaper.

Triad, HackGT 12 September 2025

- Built a web app in which two AI agents converse with each other and the user, with calendar-tracked notes (TypeScript, OpenAI, Firebase).

NeoMental, Mental Health App July 2025 – March 2026

- Prototyped an app with an emotions diary and long-term mental health resources (Swift, SQL).

OTHER EXPERIENCE

RAE Architect Malacca, Malaysia
Intern Summer 2024

- Wrote a Python script that automated a workflow procedure, cutting processing time by 8%.
- Coordinated documentation for hospital development planning; drafted contracts and joined client meetings.

SKILLS

Programming	Python, C++, C, Java, R, Swift, TypeScript, SQL
ML & Tools	PyTorch, TensorFlow, scikit-learn, NumPy, Git
Languages	English (native), Chinese (native), Malay (fluent); IELTS 8.0 overall (July 2024)